Fish Table Game Design Spec

October 17, 2026

1. Overview

A custom 10-seat fish table game for home use, points only, built to keep friends coming back. The bank targets 97–98% payback over a week.

- Two weapons: the gun (fast, shared, thrill side, same odds for everyone) and the fishing pole (personal, strategic, survival and income).
- 1 coin = 100 points.
- The one balance rule everywhere: chance × average pay = payback × cost. Change one side and the other follows.
- Moods, seesaws, jars and scenes only change when things pay, never the average.
- The director picks scenes and timing; it never changes a shot or cast after it happens.

2. Controls and table layout

Control	Gun	Pole
Joystick	Aim	Aim the cast
Double push forward	Lock on to nearest fish; left/right switches target; double forward again releases. Free, no odds change	—
Double pull back	Switch gun / pole	Switch pole / gun
Shoot	Fire	Cast
Enhance	Bet 1–100 per shot	—
Type	—	Cycle bait; press again to add a second bait (combo)
Up / down during a cast	—	Shrink or loosen the scent circle

- No weak spots. Bullets bounce off walls and off fishing bubbles.
- Fire rate starting point: about 1.5 shots/sec.
- Screen layers: the pond (center, never moves), the frame (sketch paper edges or visor), the world outside (slow, dim, at the corners). A cast gives a local half-second slow-mo ring on that seat only.

3. Gun: groups, ranges and catch chances

Payout = multiplier × bet. Catch chance per shot = group payback ÷ average multiplier. Same odds for every player at every bet; no boosts, no bet checks.

Group	Payback	Fish	Range	Avg	Catch
Tiny	98%	Sardine	2–3x	2.4x	41%
Tiny	98%	Anchovy	2–5x	3.3x	30%
Tiny	98%	Clownfish	3–5x	3.8x	26%
Tiny	98%	Butterflyfish	3–6x	4.2x	23%
Small	98%	Minnow	2–10x	4x	24.5%
Small	98%	Angelfish	3–8x	4.8x	20%
Small	98%	Pufferfish	4–7x	5.2x	19%
Small	98%	Lionfish	5–10x	7x	14%

Small	98%	Seahorse	6–15x	9.5x	10.3%
Medium	97%	Tuna	8–15x	11x	8.8%
Medium	97%	Barracuda	10–25x	16x	6.1%
Medium	97%	Octopus	12–30x	19x	5.1%
Medium	97%	Swordfish	15–35x	23x	4.2%
Medium	97%	Turtle	20–50x	32x	3.0%
Big	95%	Hammerhead	80–150x	108x	0.88%
Big	95%	Shark	100–400x	200x	0.48%
Big	95%	Manta ray	150–300x	210x	0.45%
Big	95%	Marlin	150–500x	280x	0.34%
Boss	90%	Healer Jellyfish	300–500x	380x	0.24%
Boss	90%	Megalodon	400–800x	560x	0.16%
Boss	90%	Giant Squid	500–1,000x	700x	0.13%
Boss	90%	Kraken	800–1,200x	960x	0.094%

- Moods: Stable (~60% of the time; pays near the low end, higher catch) and Swingy (~40%; pays up to the top, lower catch). Same average either way. Each group can switch on its own clock every 1–3 min.
- Wild fish: hot pays ~3x more and is ~3x harder; cold is the reverse.
- Sardine ↔ Minnow seesaw: Sardine cold 25% (36% catch, ~86%), even 50% (~98%), Sardine hot 25% (46% catch, ~110%); Minnow mirrors.
- Golden Sardine: normal Sardine 38% catch (~91%) plus ~1 in 700 shots at Sardines becomes a 50x Golden Sardine (+7%) = ~98%.
- Specials: Bomb Crab (20–150x total), Lightning Eel (30–200x), Treasure Turtle (50–300x), golden variants (2x). Chance = payback ÷ average total multiplier.
- Hot patterns are kept short and hidden (15–30 sec) or capped near 105% so players can't time big bets into them.
- Thrill targets (20-min session): 10–30x about every minute, 100x+ about every 5 min, 300x+ about once a session, 1,000x rarely.

4. Gun balance engine (the director)

Layer	Length	Controls
Pattern / scene	15–90 sec	Hot, Dry, Tease, Spotlight, Bait Ball
Wave	2–5 min (avg ~3)	Shape: Calm, One spike, Rollercoaster, Tension build
Arc	3–6 waves	Boss Saga, Rising Tide, Journey, Comeback, Season Opener
Cycle	~30 min	Short-term ledger
Block	3 hours	Main balance target
Week	—	Final target: 97–98%

- Per wave (10 players, avg bet 8, 1.5 shots/sec, 70% active): ~15,000 points wagered; bank keeps ~3% on average; short stretches swing hard.
- Risk budget: about −3,000 per wave, −10,000 per cycle, back on target by end of each 3-hour block. Any shortfall carries forward; never written off.
- Recovery is gentle (calm, hunt, boss scenes). Floor ~92% for any scene.
- Surplus is spent on fun (Bait Balls, spotlights, hot seesaws).
- Experiments: 10–15% of waves are wild tries (All Giants, Double Boss, Tiny Storm, Silent Recovery, Surprise Spike) with fixed payback; stop early if people leave or the bank drops fast.
- Long game (needs weeks of logs): generous openings, save for the weekend, comeback nights, multi-day events, bigger risk budget once variance is known.
- Decision rhythm: every shot is logged; the director re-evaluates every 2–5 sec and at scene ends.

- Settings sliders: volatility, chaos frequency, boss rhythm, arc length, wildcard rate, predator aggression, biome mix, tells on/off, movement speed, pattern variety, bank target, risk budget.
- Engagement signals: seats filled/joining/leaving, shots per minute, bet changes, idle seats, session length, return visits. Train on return visits as well as session length.

5. Scenes, roles and movement

Waves are built from premade scenes (20–40 sec) assembled live. Example wave: Morning Reef (~99%) → The Hunt (~96%) → Bait Ball (~108%) → Boss (~90%) ≈ 96%.

Role	Fish	Behavior
Swarm	Sardine, Anchovy	Big fast schools, flocking
Reef	Clownfish, Butterflyfish	Slow drifting loops
Runner	Minnow	Zigzag
Glow	Angelfish	Golden flashes
Blocker	Pufferfish	Soaks bullets
Escort	Lionfish	Guards big fish
Drifter	Seahorse	Rare, slow
Hunters	Barracuda, Swordfish	Chase swarms
Cruisers	Tuna, Turtle, Octopus	Evolving sine waves, figure-8s
Giants	Big fish, boss	Grand paths, U-turns

- Temperaments: Anchors (Sardine, Clownfish, Tuna, Turtle, Hammerhead), Swingers (Minnow, Angelfish, Barracuda, Shark, Marlin), Wildcards (specials).
- Group rules (same range): seesaw, shared mood, spotlight, golden variant.
- Cross-group rules: escort, predator chase, chain, jar link.
- Regulars: Ghost Marlin (returns bigger each wave up to 3, same average), Lucky Seven (catch all 7 for a 30x bonus), Old Turtle.
- Tells: lantern fish → Night Water; bird shadows → Bait Ball; remoras → Shark; bubbles from the bottom → Boss.
- Linked fish: mates (catch one, its mate panics and pays double briefly), rivals (Barracuda vs Swordfish fight; winner pays more), followers (catching them slows the big fish).
- One wildcard slot per scene (Golden fish, Treasure Turtle, Bomb Crab, Lightning Eel, Mystery Egg), max once per wave.
- Per wave (~3 min): Tiny 210, Small 190, Medium 90, Big 30, Boss 1 (~521 total, ~37 on screen).
- Spawn in schools: Tiny 5–8 every 5–6 sec; Small 3–6 every 4–5 sec; Medium 1–3 every ~4 sec; Big 1 (+escorts) every ~6 sec.
- Crossing times: Tiny/Small 8–12 sec; Medium 15–20 sec; Big 20–30 sec; boss 60–90 sec. Next group enters when the current one is ~2/3 across. Max 2 big fish on screen.
- Movement types: fixed signature formations, adaptive behaviors (flocking, flee, chase, bait pull, avoid bubbles), evolving loops that change every 2–3 waves (speed, side, direction, height, size).
- Pattern library: straight, sine, zigzag, circle, figure-8, spiral, V-formation, curtain, scatter burst, U-turn, dive and rise.
- Boss fix: the boss is caught only ~1 in 10 waves at 5% of shots; use focus fire during boss waves, a longer stay, or a team-raid health bar.

6. Fishing pole: baits and fish

One bite per cast. “Bite” = the chance a fish takes the bait and is landed when it swims into the circle.

Bait	Fish	Style	Bite	Pays	Payback
Worms (15)	Sardine	Moody	see below	15 → 130	~105%
Worms (15)	Minnow	Steady	40%	35 / 38 / 40	~100%

Worms (15)	Anchovy	Steady	25%	55 / 60 / 65	~100%
Shrimp (30)	Clownfish	Moody	see below	30 → 260	~100%
Shrimp (30)	Angelfish	Steady + Golden	35%	70 / 75 / 80; 1 in 20 Golden 300	~100%
Shrimp (30)	Butterflyfish	Steady	25%	110 / 120 / 130	~98%
Squid (60)	Pufferfish	Big swings	see below	60 → 520	~97%
Squid (60)	Lionfish	All or nothing	20%	290 flat	~97%
Crab (80)	Tuna	Moody, sizes	see below	80 → ~450	~97%
Crab (80)	Turtle	Steady	30%	250 / 260 / 270	~97%
Live bait (120)	Barracuda	Hustler	60%	170 / 190 / 210	~93%
Live bait (120)	Swordfish	The Price	25.5%	500 / 600 / 750 / 900; 1 in 5 escapes	~95%
Mackerel (160)	Hammerhead	Steady big	15%	900 / 950 / 1,000	~89%
Chum (200)	Shark	Frenzy moods	see below	350 → 1,600	~89%
Chum (200)	Manta ray	Gentle giant	30%	580 / 600 / 620	~90%
Golden lure (500)	Golden Koi	Pocket	8.8%	5,000	~88%

Three-amount pays split 40 / 40 / 20%. Ladders split 20 / 30 / 30 / 10 / 10% unless noted.

Moods:

Fish	Low (busy)	Normal	Wild
Sardine	25% of time; 80% bite; 15 (85%) / 30 (15%)	50%; 58%; 15/20/30/40/50	25%; 26%; 45/55/70/90/130
Clownfish	25%; 77%; 30 (85%) / 60 (15%)	50%; 56%; 30/40/60/80/100	25%; 24.5%; 90/110/140/180/260
Pufferfish	40%; 75%; 60 (90%) / 120 (10%)	20%; 54%; 60/80/120/160/200	40%; 24.5%; 180/220/280/360/520
Tuna	30%; 72%; 80 (85%) / 160 (15%)	40%; 39%; 20–60 lb → ~200	30%; 19.5%; 60–120 lb → ~450
Shark	—	75%; 12%; 1,400 / 1,500 / 1,600	Frenzy 25%; 45%; 350 / 400 / 450

- Moods switch quietly every 15–30 sec in no set order.
- Jars: each moody fish has its own jar. Low casts feed it (~2 points per Worm cast); Wild only turns on when the jar can pay for it.
- Squid is the crossroads: players either level up to medium bait or drop back down.
- Risk is the balance, not the odds: keep ~20 casts' worth of balance (Worms ~300, Shrimp ~600, Squid ~1,200).

Exceptions (only cross-bait bites): Clownfish on Worms, Pufferfish on Shrimp, Tuna on Squid. A fish won't bite bait worth more than it pays.

Combos (two baits on one hook; cost = both):

Combo	Fish	Bite	Pays	Payback
Worms + Shrimp (45)	Seahorse (combo only)	30%	140 / 150 / 160	~100%
Worms + Shrimp (45)	Angelfish	53%	70–80; 1 in 20 Golden 300	~100%
Worms + Shrimp (45)	Butterflyfish	38%	110 / 120 / 130	~100%
Squid + Crab (140)	Octopus (combo only)	28%	460 / 480 / 500	~96%

Squid + Crab (140)	Lionfish	46%	290 flat	~96%
Squid + Crab (140)	Turtle	52%	250 / 260 / 270	~96%
Shrimp + Crab (110)	Swordfish	23%	500–900; 1 in 5 escapes	~95%
Live bait + Mackerel (280)	Marlin (combo only)	11.7%	2,000 / 2,200 / 2,400	~92%
Live bait + Mackerel (280)	Hammerhead	27%	900 / 950 / 1,000	~92%
Chum + Golden (700)	Wave boss	4%	~15,000	~85%

Combos scare off small fish. Unlisted pairs act like the more expensive bait.

7. Fishing systems

- Scent circle radius: base ~150 px, medium ~200, big ~250, golden ~300. Long Rod adds 30%.
- Circle overlap: under 25% nothing special; 25–60% combo zone (even across players); over 60% crowded, bites drop unless it's a combo pair.
- Repel rules: Chum scares small fish; Golden lure scares everything except the Koi; Squid next to Worms scares Sardines.
- Fight: fish icon flashes green / yellow. Press on green, let go on yellow. Miss a green or press on yellow and it escapes. Greens needed: small 2, steady 3, big hitter 4, medium 4, big 5, Koi/boss 6. Several fish fight one at a time, smallest first. Skill nudges landing about ±5%.
- Bait spoils after 30 sec. The timer pauses while bullets spook the fish.
- Bubbles: bullets bounce off a fishing bubble; fish inside are safe.
- Rods (pick one per session): Swift beats Fast fish, struggles with Heavy; Heavy beats Heavy, struggles with Tricky; Flex beats Tricky, struggles with Fast. In shared circles a fish picks a line 3 : 2 : 1 by rod match. Matched rod: greens 20% wider; wrong rod: 20% narrower. Rods never change bite chances.
- Fish types: Fast (Sardine, Anchovy, Butterflyfish, Barracuda, Swordfish, Marlin, Megalodon), Heavy (Pufferfish, Tuna, Turtle, Hammerhead, Shark, Kraken), Tricky (Minnow, Clownfish, Angelfish, Lionfish, Seahorse, Octopus, Manta ray, Giant Squid, Healer Jellyfish).
- Fisher upgrades: Long Rod (bigger circle) and Bait Saver (escaped fish don't eat the bait).
- Golden lure pocket (hidden): fed by 2% of medium bait, 4% of big bait and every lure bought. Koi odds about 1 in 11 normally, about 1 in 20 if the pocket is under 5,000, about 1 in 9 if over 15,000. Pays 5,000 from the pocket.
- Inventory and market: catches go to inventory; sell any time. Sell price changes by season demand: Hot +25%, Normal, Low −20%. Fish spoil after 2 seasons. Base values sit about 5% lower to pay for patient sellers.
- Selling bonuses are off for now. Later version: rod bonus +3%, weight bonus only for heavy fish, no minimum per fish.
- During dream drops, fishing pauses: lines reel in, bait timers freeze.

8. World

- Lore: the world is a digital ocean that's glitching. Fishing is survival and income; the gun is a stabilizer that locks glitched creatures back into place and turns them into points.
- Seasons (2–4 waves each): Winter (Japan, Hokkaido; calm, Heavy rod season, Tuna in demand), Monsoon (Bay of Bengal; best fishing season, about +3% for fishing, funded by the gun running cooler), Summer reef (Australia; lively, more golden fish, reef fish in demand).
- The background stays the same ocean; seasons show through overlays (light, weather particles, colors, music).
- Captain Rio flies a seaplane between stops; 8–12 sec cutscenes play over the ocean while fish keep swimming. Guest pilots sometimes take other routes. Sometimes a transition turns into a mission (bonus round with a fixed payback).
- Sketch Man: the ocean is his drawing; his hands and paper edges show at the corners. Scene changes are page turns (pencil lines, then color fills in). Glitches are ink smudges.

- Dream drops (rare, short): On top of the TV, Kingdom Gates → Space (the king turns into an alien), Cloud Kingdom, bathtub, kid's aquarium, desert mirage, lava lamp.
- Dream drop rules: same sizes and hitboxes, value ring on every creature, paths fair to all 10 seats, calm background under the fish, same odds, 2–3 sec teleport.
- Toy box: every creature has fixed stats wherever it appears. New creatures copy an ocean fish's stats (UFO = Octopus, Rubber Duck = Clownfish, Wind-up Fish = Pufferfish, Comet = Marlin, Alien King = Kraken).
- Value rings: gray tiny, green small, blue medium, gold big, red boss. Glowing trail = hot/wild; sparkle trail = golden.
- Art hand-off: PNG with transparency or SVG, facing right, one consistent view; sizes roughly tiny 32 px, small 48, medium 80, big 160, boss 320+; ring drawn by the game; 4–8 frame swim loops; names like medium_ufo_01.png.

9. Algorithms

Gun, per shot:

```
on_shot(seat, bet):
  balance[seat] -= bet
  fish = first creature the bullet hits (lock-on just aims)
  P = group payback adjusted by current state (states average to group P)
  avg_m = average multiplier for this fish in the current mood
  chance = P / avg_m
  if random() < chance:
      m = draw multiplier from fish's range for this mood
      pay seat m × bet; remove fish
  log(seat, bet, fish, hit, pay)
```

Director, every 2–5 sec and at scene ends:

```
ledger = total_wagered × target − total_paid
if ledger < risk floor    → recovery shapes (Calm, Hunt, Boss)
if ledger > comfort level → spikes, Bait Balls, spotlights
read table energy → choose next scene and dials
10–15% of choices are experiments with fixed payback
hot states only turn on when their jar can pay
settle at wave → cycle → block → week
```

Fishing, per cast:

```
on_cast(seat, bait):
  balance[seat] -= bait cost
  radius = bait size (+30% Long Rod)
  for each fish entering the circle:
      skip if it doesn't eat this bait
      if random() < bite(fish, bait, mood):
          run the green/yellow fight (skill ±5%)
          pay = draw from the fish's pay table
          Swordfish: 1 in 5 escapes → 0
          add to inventory; end the cast
  bait spoils at 30 sec (paused while spooked)
```

Logging: every shot and cast (seat, bet or bait, fish, hit or miss, pay), every wave (shape, scenes, in/out), every seat (session length, start/end balance, style), every block (payback vs target), and engagement signals.

10. Open items and next steps

1. Build the simulator: thousands of fake sessions across settings to confirm the bank lands on target.
2. Godot prototype: one screen, a few creatures, one seat.
3. Boss raid design (health bar, phases, assist rewards).

4. Net / laser / technologies (Fish Train) are parked.
5. Hardware: I/O board model number still needed; power supply short not yet found.
6. Train the director from logs once real play data exists.